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## Filter element and filter system


### Abstract

A filter element has a filter medium arranged between first and second end disks. The first end disk has a positioning and sealing section having external positioning recesses for engaging disturbance geometries of a filter housing with form fit to position the filter element circumferentially in the filter housing. The positioning and sealing section has an inner interface for radially sealing and axially positioning the filter element relative to the filter housing. The interface, viewed along a longitudinal direction from first toward second end disk, extends farther into the positioning and sealing section than the positioning recesses. The interface has an annular seal groove surrounding a symmetry axis of the filter element and configured to engage a seal rib of the filter housing with form fit for axial positioning relative to the filter housing. A filter system is provided with such a filter element in its filter housing.



**Inventors:** **Stark; Dennis (Mauer, DE), Kaufmann; Michael (Meckenheim, DE), Ruhland; Klaus-Dieter (Meckenheim, DE)**

**Applicant:** **MANN+HUMMEL GmbH** (Ludwigsburg, DE)



**Family ID:** **1000008819888**

**Assignee:** **MANN+HUMMEL GmbH (Ludwigsburg, DE)**

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*Primary Examiner:* He; Qianping

## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of international application No. PCT/EP2021/073281 having an international filing date of 23 Aug. 2021 and designating the United States, the international application claiming a priority date of 24 Aug. 2020 based on prior filed German patent application No. 10 2020 122 024.3, the entire contents of the aforesaid international application and the aforesaid German patent application being incorporated herein by reference.

BACKGROUND OF THE INVENTION
(1) The present invention concerns a filter element and a filter system with such a filter element.
(2) A filter system, for example, an air filter system, comprises a filter housing and a filter element received in the filter housing so as to be removable. Such a filter element can comprise a folded filter medium which is placed between two end disks and is fixedly connected to them. At least one of the end disks can comprise a seal element with which the filter element can be sealed in relation to the filter housing. Furthermore, this end disk can also have a contour in which a corresponding counter contour of the filter housing can engage with form fit. In this way, it is ensured that only filter elements matching the filter system can be installed in the filter housing. This means filter elements without such a matching contour cannot be installed in the filter housing.
(3) WO 2012/172019 A1 shows an air filter system comprising a housing with a housing top part and an air filter element for filtering air. The housing comprises a clean air socket for discharge of clean air from the housing and a seal receptacle for form fit connection to a cylindrical seal of the

air filter element and for holding the air filter element, wherein the seal receptacle comprises a cylindrical seal surface extending from the housing top part into the interior of the housing, the seal surface enclosing the clean air socket and the seal of the air filter element contacting it radially, wherein at the seal surface an annular collar which is projecting radially past the seal surface is arranged which can be gripped with form fit by the seal of the air filter element.

SUMMARY OF THE INVENTION

(4) In view of this background, the present invention has the object to provide an improved filter element.

(5) Accordingly, a filter element for a filter system is proposed. The filter element comprises a filter medium, a first end disk connected to the filter medium, and a second end disk connected to the filter medium, wherein the filter medium is arranged between the first end disk and the second end disk, wherein the first end disk comprises a positioning and sealing section facing away from the filter medium, wherein the positioning and sealing section comprises external positioning recesses which are configured such that disturbance geometries of a filter housing of the filter system engage with form fit the positioning recesses in order to position the filter element circumferentially in relation to the filter housing, wherein the positioning and sealing section comprises at the inner side an interface which is configured to seal the filter element radially in relation to the filter housing and to position it axially in relation to the filter housing, wherein the interface, viewed along a longitudinal direction of the filter element which is oriented from the first end disk in the direction toward the second end disk, extends farther into the positioning and sealing section than the positioning recesses, wherein the interface comprises a seal groove in the form of an annular groove which extends circumferentially completely about a symmetry axis of the filter element, and wherein the seal groove is configured such that a seal rib of the filter housing engages with form fit the seal groove in order to position the filter element axially in relation to the filter housing.

(6) Because the interface extends farther into the positioning and sealing section than the positioning recesses, it can be achieved that an interface of the filter housing which is projecting past the disturbance geometries in the longitudinal direction can be guided at the interface of the filter element. In this way, the filter element is guided at the filter housing when the filter element is rotated relative to the filter housing in order to position it in relation to the filter housing.

(7) The filter element is preferably an air filter element. The filter element is in particular suitable for purifying air supplied to an air compressor. The filter medium is preferably folded. The filter medium forms in particular a circumferentially closed folded bellows. The end disks are preferably manufactured of a plastic material. For example, the end disks are foamed or cast onto the filter medium at the end face. Preferably, the first end disk as well as the second end disk are fixedly connected to the filter medium. The filter element is preferably constructed with rotational symmetry in relation to a center or symmetry axis. Also, the end disks are in particular constructed with rotational symmetry in relation to this symmetry axis. However, the filter element can also have an oval geometry in cross section. In this case, the filter element is not constructed with rotational symmetry in relation to the symmetry axis.

(8) As an alternative, the positioning and sealing section can be provided also at the second end disk. The positioning and sealing section is configured as one piece, in particular monolithic as one piece, together with the first end disk. “One piece” or “one part” means presently that the first end disk and the positioning and sealing section form a common component and are not assembled of different components. “Monolithic as one piece” means presently that the first end disk and the positioning and sealing section are manufactured throughout of the same material.

(9) The positioning and sealing section extends at the end face away from an end surface of the first end disk which is facing away from the filter medium. The positioning and sealing section is constructed with rotational symmetry in relation to the symmetry axis and extends circumferentially completely around it. The positioning and sealing section is thus preferably of an

annular shape. The number of the positioning recesses is arbitrary. That the positioning recesses are "externally" arranged at the positioning and sealing section means presently that the positioning recesses face away from the symmetry axis.

(10) A form fit connection is produced by two connection partners, presently the disturbance geometries of the filter housing and the positioning recesses of the filter element, mutually engaging each other or engaging from behind. In this context, the disturbance geometries, in a mounted state of the filter element, engage with form fit the positioning recesses so that the filter element cannot be rotated relative to the filter housing. As long as these disturbance geometries do not yet engage the positioning recesses, it is possible to rotate the filter element in relation to the filter housing. The filter element can be rotated in relation to the filter housing until the positioning recesses are aligned with the corresponding disturbance geometries so that the disturbance geometries engage the positioning recesses.

(11) As soon as the disturbance geometries engage in the positioning recesses, the filter element can be inserted in an insertion direction, which is preferably opposite to the longitudinal direction of the filter element, into the filter housing. In this way, the interface of the filter element seals in relation to the filter housing. The filter housing comprises preferably an interface which corresponds to the interface of the filter element. That the positioning recesses are suitable to position the filter element "circumferentially" in relation to the filter housing means therefore presently that a rotation of the filter element in relation to the filter housing is no longer possible when the disturbance geometries engage the positioning recesses.

(12) That the interface is arranged "at the inner side" at the positioning and sealing section means presently that the interface is arranged so as to face the symmetry axis. Preferably, the interface is cylinder-shaped. The interface extends preferably completely around the symmetry axis. The interface is at least in sections resiliently deformable so that it can be radially compressed in relation to the filter housing. "Radial" means in this context along a radial direction oriented perpendicularly to the symmetry axis and facing away from it. For sealing the filter element in relation to the filter housing, the interface at least in sections is radially compressed and resiliently deformed thereby.

(13) That the interface is suitable for positioning the filter element "axially" in relation to the filter housing is to be understood presently in particular such that a fixation or positioning of the filter element along the longitudinal direction or along the insertion direction is possible by means of the interface. For this purpose, the filter housing can engage the interface, for example, with form fit. That the interface, viewed along the longitudinal direction, extends farther into the positioning and sealing section than the positioning recesses means presently that the interface, in relation to a circumferentially extending end surface of the positioning and sealing section, comprises a larger depth than the positioning recesses.

(14) In embodiments, the interface comprises a circumferentially extending sealing surface which is configured to seal the filter element radially in relation to the filter housing, wherein the seal groove, viewed along the longitudinal direction, is arranged behind the seal surface. This means that the seal rib engages the seal groove not until the seal surface of the filter element is compressed relative to a seal surface of the filter housing. The seal surface comprises preferably a circular cylindrical geometry which extends circumferentially completely around the symmetry axis. When radially sealing the filter element in relation to the filter housing, the seal surface is compressed so that the positioning and sealing section is deformed, at least in sections. The seal groove is in particular an annular groove which extend circumferentially completely around the symmetry axis. The seal groove, viewed along the longitudinal direction, is arranged adjacent to the seal surface. This means that the seal groove and the seal surface are positioned adjacent to each other.

(15) In embodiments, the interface comprises a circumferentially extending surface which is positioned such that the seal groove, viewed along the longitudinal direction, is arranged between

the seal surface and the surface, wherein the surface, viewed along the longitudinal direction, extends farther into the positioning and sealing section than the positioning recesses. The surface extends circumferentially completely around the symmetry axis and comprises preferably a circular cylindrical geometry. In a mounted state of the filter element, the surface is preferably not in contact with the filter housing. This means that between the filter housing and the surface a gap is provided, in particular an air gap.

(16) In embodiments, the surface comprises a larger diameter than the seal surface. This means in particular that the surface, viewed in the radial direction, extends farther into the positioning and sealing section than the seal surface.

(17) In embodiments, the positioning recesses extend, viewed along the longitudinal direction, farther into the positioning and sealing section than the seal surface. This means that the positioning recesses, viewed along the longitudinal direction, comprise a larger depth than the seal surface. In other words, the seal surface ends, viewed along the longitudinal direction, in front of the positioning recesses.

(18) In embodiments, the first end disk is open and the second end disk is closed. In reverse, also the first end disk can be closed and the second end disk open. Furthermore, also both end disks can be open. “Open” means presently that the respective end disk has a passage, in particular an outflow opening, through which a fluid, for example air, can flow out of an interior of the filter element. The passage is preferably constructed with rotational symmetry in relation to the symmetry axis.

(19) In embodiments, the positioning recesses are arranged distributed uniformly or non-uniformly about a circumference of the filter element. In particular, the positioning recesses are arranged uniformly or non-uniformly distributed about the symmetry axis. “Uniformly” means presently that circumferential angles between the individual positioning recesses or distances between the individual positioning recesses are constant or identical. “Non-uniformly” means presently that circumferential angles between the individual positioning recesses or distances between the individual positioning recesses are differently sized. The number of positioning recesses is arbitrary. For example, at least three positioning recesses are provided. However, also four positioning recesses, five positioning recesses, six positioning recesses or more than six positioning recesses can be provided.

(20) Furthermore, a filter system with a filter housing and such a filter element is proposed. In this context, the filter housing comprises disturbance geometries which engage with form fit the positioning recesses of the filter element, wherein the filter housing comprises a seal surface with which the interface of the filter element, at least in sections, is radially compressed, and wherein a respective end surface of the disturbance geometries, viewed along the longitudinal direction, is arranged behind the seal surface of the filter housing.

(21) This means the end surface of the disturbance geometry is leading relative to the seal surface of the filter housing. In other words, the filter element, upon installation in the filter housing, first contacts the end surface of the disturbance geometries prior to the interface of the filter element sealing relative to the seal surface of the filter housing. The filter housing comprises preferably a housing bottom part as well as a housing top part which can be separated from each other in order to exchange the filter element. The housing bottom part is in particular cup-shaped and comprises a bottom section as well as a hollow cylindrical base section connected to the bottom section so as to form one piece. The interface of the housing bottom part is provided at the bottom section and comprises the seal surface of the filter housing. The interface of the filter element is suitable for interacting with the interface of the filter housing such that the filter element is sealed in relation to the filter housing and positioned relative thereto. Also, the disturbance geometries are provided preferably at the bottom section. The housing bottom part comprises furthermore a laterally arranged fluid inlet as well as a fluid outlet arranged at the bottom section. The fluid inlet is preferably arranged perpendicularly to the symmetry axis. The fluid outlet, on the other hand, is

preferably constructed with rotational symmetry in relation to the symmetry axis.

(22) In embodiments, the positioning and sealing section of the filter element, upon installation thereof in the filter housing, contacts with the end face first the disturbance geometries, wherein the disturbance geometries by means of a rotation of the filter element in relation to the filter housing can be brought into form fit engagement with the positioning recesses. This means that the filter element is first inserted in the insertion direction into the filter housing, in particular the filter housing bottom part, until the positioning and sealing section rests on the disturbance geometries. By a rotation of the filter element in relation to the filter housing, the positioning recesses now can come to engage the disturbance geometries so that the filter element can be inserted farther into the housing bottom part in order to seal the filter element in relation to the filter housing.

(23) In embodiments, a seal surface of the interface of the filter element, upon installation of the filter element in the filter housing, contacts the seal surface of the filter housing not until the disturbance geometries are in form fit engagement with the positioning recesses. This means that the filter element, as mentioned before, can be inserted farther into the filter housing only when the disturbance geometries are aligned with the positioning recesses.

(24) In embodiments, the disturbance geometries comprise, viewed along the longitudinal direction, a larger depth than the seal surface of the interface. The depth is measured in particular as a distance of an end surface of the positioning and sealing section in relation to an end surface of the first end disk.

(25) In embodiments, the filter housing comprises a seal rib which engages with form fit a seal groove of the interface of the filter element, wherein the seal rib, viewed along the longitudinal direction, is arranged behind the seal surface of the filter housing. The seal rib is annular and extends circumferentially completely around the symmetry axis. Since the seal rib, viewed along the longitudinal direction, is arranged behind the seal surface of the filter housing, the seal rib engages the seal groove not until the seal surface of the interface of the filter element contacts the seal surface of the interface of the filter housing and is radially compressed relative thereto.

(26) In embodiments, the filter housing comprises a centering surface, wherein the centering surface, viewed along the longitudinal direction, is arranged behind the seal rib of the filter housing. The centering surface is preferably circular cylindrical and extends circumferentially completely around the symmetry axis. At the centering surface, the seal surface of the interface of the filter element is guided as long as the disturbance geometries have not yet engaged the positioning recesses of the positioning and sealing section of the filter element, i.e., as long as the positioning and sealing section is resting with the end face on the disturbance geometries and the filter element is rotatable relative to the filter housing.

(27) In embodiments, between the centering surface of the filter housing and the interface of the filter element, a circumferential gap is provided in an installed state of the filter element in the filter housing. The gap is in particular an air gap. In the installed state of the filter element, the disturbance geometries engage with form fit the positioning recesses so that the seal surface of the filter element is radially compressed with the seal surface of the filter housing.

(28) In embodiments, the disturbance geometries are arranged distributed uniformly or non-uniformly about a circumference of the filter housing. The number of disturbance geometries is arbitrary. For example, three or five such disturbance geometries are provided. “Uniformly” means presently that a circumferential angle or distance between the individual disturbance geometries is constant. “Non-uniformly” means presently that the circumferential angle or distance between the individual disturbance geometries is non-uniform.

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## Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. **1** shows a schematic plan view of an embodiment of a filter system.

(2) FIG. **2** shows a further schematic plan view of the filter system according to FIG. **1**.

(3) FIG. **3** shows a schematic section view of the filter system according to the section line III-III of FIG. **1**.

(4) FIG. **4** shows a further schematic section view of the filter system according to the section line IV-IV of FIG. **2**.

(5) FIG. **5** shows a further schematic section view of the filter system according to the section line IV-IV of FIG. **2**.

(6) FIG. **6** shows a further schematic section view of the filter system according to the section line IV-IV of FIG. **2**.

(7) FIG. **7** shows a detail view VII according to FIG. **3**.

(8) FIG. **8** shows a detail view IIX according to FIG. **6**.

(9) FIG. **9** shows a schematic perspective view of an embodiment of a filter housing for the filter system according to FIG. **1**.

(10) FIG. **10** shows a schematic perspective view of an embodiment of a filter element for the filter system according to FIG. **1**.

(11) FIG. **11** shows a schematic side view of the filter element according to FIG. **10**.

(12) In the Figures, same or functionally the same elements, if nothing to the contrary is indicated, are provided with the same reference characters.

DETAILED DESCRIPTION OF THE INVENTION

(13) FIG. **1** shows a schematic plan view of an embodiment of a filter system **1**. FIG. **2** shows a further schematic plan view of the filter system **1**. FIG. **3** shows a schematic section view of the filter system **1** according to the section line III-III of FIG. **1**. FIG. **4** shows a further schematic section view of the filter system **1** according to the section line IV-IV of FIG. **2**. FIG. **5** shows a further schematic section view of the filter system **1** according to the section line IV-IV of FIG. **2**. FIG. **6** shows a further schematic section view of the filter system **1** according to the section line IV-IV of FIG. **2**. FIG. **7** shows the detail view VII according to FIG. **3**. FIG. **8** shows the detail view IIX according to FIG. **6**. FIG. **9** shows a schematic perspective view of an embodiment of a filter housing **2** for the filter system **1**. FIG. **10** shows a schematic perspective view of an embodiment of a filter element **3** for the filter system **1**. FIG. **11** shows a schematic side view of the filter element **3**. In the following, reference is being had to FIGS. **1** through **11** at the same time.

(14) The filter system **1** can also be referred to as filter assembly. The filter system **1** is used preferably as intake air filter for air compressors. Alternatively, the filter system **1** can however be used also as intake air filter for internal combustion engines, for example, in motor vehicles, trucks, construction vehicles, watercraft, rail vehicles, agricultural machines or vehicles, or in aircraft. The filter system **1** can also be used in immobile applications, for example, in the building technology. The filter element **3** is suitable in particular for filtering intake air of an air compressor. Preferably, the filter element **3** is an air filter element.

(15) The filter element **3** is constructed with rotational symmetry in relation to a center or symmetry axis **4**. The filter element **3** comprises a filter medium **5** which is cylinder-shaped. The filter medium **5** is constructed with rotational symmetry in relation to the symmetry axis **4**. For example, the filter medium **5** can be of a closed annular shape and can be present in the form of a folded bellows folded in a star shape. The filter medium **5** is thus preferably folded.

(16) The folded filter medium **5** can be provided with a stabilization ring **6** for stabilization thereof. The stabilization ring **6** can also be referred to as fixation coil. The stabilization ring **6** is, for example, a strip glued onto the filter medium **5** or a glued-on string. The stabilization ring **6** can be an adhesive bead or glue bead or the like, extending circumferentially completely around the symmetry axis **4** about the filter medium **5**. In particular, the stabilization ring **6** can comprise a hot

melt and/or hot melt-impregnated threads, for example, at least three such threads. The stabilization ring **6** serves for stabilizing the folds of the folded filter medium **5** and to thus keep their distance relative to each other identical. The stabilization ring **6**, viewed along the longitudinal direction LR of the filter element **3**, is positioned off-center at the filter medium **5**.

(17) In this context, the longitudinal direction LR is oriented along the symmetry axis **4**. In the orientation of FIG. **11**, the longitudinal direction LR can be oriented from bottom to top. The longitudinal direction LR can however also be oriented in reverse. The stabilization ring **6** in this context is provided at the exterior at the filter medium **5**. “Off-center” means presently that the stabilization ring **6** in relation to a first end face **7** and a second end face **8** of the folded filter medium **5** is not centrally arranged between the two end faces **7**, **8** but, for example, closer to the first end face **7** than to the second end face **8**. In particular, precisely one annular stabilization ring **6**, extending circumferentially completely around the symmetry axis **4**, is provided.

(18) The filter medium **5** is, for example, a filter paper, a filter fabric, a laid filter or a filter nonwoven. In particular, the filter medium **5** can be produced by a spun-bond or melt-blown method or can comprise such a fiber layer applied onto a nonwoven or cellulose support. Furthermore, the filter medium **5** can also be felted or needled. The filter medium **5** can comprise natural fibers, such as cellulose or cotton, or synthetic fibers, for example, of polyester, polyvinyl sulfite or polytetrafluoroethylene. During processing, fibers of the filter medium **5** can be oriented in, at a slant to and/or transversely to or randomly in relation to a machine direction.

(19) The filter element **3** comprises a first, in particular open, end disk **9** which is provided at the first end face **7** of the filter medium **5**. Moreover, the filter element **3** comprises a second, in particular closed, end disk **10** which is provided at the second end face **8** of the filter medium **5**. This means the filter medium **5** is positioned between the first end disk **9** and the second end disk **10**. The end disks **9**, **10** can be manufactured, for example, of a polyurethane material which is in particular cast in casting shells, preferably foamed. The end disks **9**, **10** can also be cast onto the filter medium **5**. The first end disk **9** is connected to the first end face **7**. The second end disk **10** is connected to the second end face **8**.

(20) The first end disk **9** comprises a centrally arranged passage **11**. The passage **11** can be an outflow opening of the filter element **3**. The first end disk **9** comprises a plate-shaped base section **12** which is connected to the first end face **7** of the filter element **3**. The passage **11** passes through the base section **12**. The exterior of the base section **12** can be provided with a plurality of grooves or cutouts **13** which are distributed uniformly around the symmetry axis **4**.

(21) Facing away from the first end face **7** of the filter medium **5**, a positioning and sealing section **14** of the first end disk **9** extending in an annular shape circumferentially around the symmetry axis **4** extends away from the base section **12**. By means of the positioning and sealing section **14**, the filter element **3** can be positioned in the filter housing **2** and sealed relative thereto, as will be explained in the following. The passage **11** passes also through the positioning and sealing section **14**.

(22) At the exterior, i.e., facing away from the passage **11**, a plurality of positioning recesses **15** are provided at the positioning and sealing section **14** of which only one is provided with a reference character in FIGS. **10** and **11**, respectively. The positioning recesses **15** are arranged distributed uniformly about the symmetry axis **4**. For example, six such positioning recesses **15** are provided. The number of the positioning recesses **15** is however arbitrary. Beginning at an annular end surface **16** of the positioning and sealing section **14**, the positioning recesses **15**, viewed along the symmetry axis **4** or along the longitudinal direction LR, comprise a depth t**15** (FIG. **8**). The positioning recesses **15** extend, beginning at the end surface **16**, in the direction toward the base section **12**.

(23) As also shown in FIG. **8**, the first end disk **9** or the positioning and sealing section **14** at the inner side, i.e., facing the passage **11**, comprises a cylindrical seal surface **17** which is constructed with rotational symmetry in relation to the symmetry axis **4** and extends circumferentially

completely around it. The seal surface **17** is suitable for interacting with the filter housing **2** in order to thus seal the first end disk **9** in relation to the filter housing **2** fluid-tightly. In this context, the seal surface **17** can be radially compressed. “Radially” means in this context in a direction of a radial direction R which is perpendicularly oriented in relation to the symmetry axis **4** and is pointing away from it.

(24) Beginning at the end surface **16**, the seal surface **17** extends along the longitudinal direction LR by a depth t**17** into the passage **11**. An annular groove or seal groove **18** extending circumferentially in a ring shape about the symmetry axis **4** adjoins the seal surface **17**. Beginning at the end surface **16**, the seal groove **18** ends at a depth t**18** along the longitudinal direction LR. In this context, the depth t**18** is smaller than the depth t**15**. The depth t**17** is smaller than the depth t**15**. Viewed along the longitudinal direction LR, a cylindrical surface **19** extending circumferentially around the symmetry axis **4** adjoins the seal groove **18**. Viewed relative to the radial direction R, the seal surface **17** comprises a smaller diameter than the surface **19**. The seal surface **17**, seal groove **18**, and the surface **19** form a seal interface or interface **20** of the filter element **3**. The interface **20** can also be referred to as first interface or as filter element interface. The interface **20** is suitable for interacting with the filter housing **2**. Beginning at the end surface **16** of the positioning and sealing section **14**, the interface **20** comprises a depth t**20**. The interface **20** can comprise also the positioning recesses **15**.

(25) Now returning to FIG. **11**, second end disk **10** comprises a plate-shaped base section **21** which is constructed with rotational symmetry in relation to the symmetry axis **4** and closes fluid-tightly the second end face **8** of the filter element **5**. Positioning elements **22** facing away from the second end face **8**, of which in FIG. **11** only one is provided with a reference character, extend away from the base section **21**. The number of positioning elements **22** is arbitrary. For example, five such positioning elements **22** can be provided which are arranged uniformly distributed around the symmetry axis **4**.

(26) The function of the filter element **3** will be explained in the following with the aid of FIG. **3**. Fluid L to be purified, for example, air, passes from a raw side RO of the filter element **3** through the filter medium **5** to a clean side RL of the filter element **3** surrounded by the filter medium **5**. This means that fluid L flows through the filter medium **5** into an interior **23** of the filter element **3** surrounded by the filter medium **5**. The purified fluid L flows out of the filter element **3** through the passage **11** of the first end disk **9** as filtered fluid L.

(27) Now returning to the filter housing **2**, the latter comprises a housing bottom part **24** and a housing top part **25**. The housing top part **25** can also be referred to as housing cover. The housing top part **25** can be removed from the housing bottom part **24** for exchanging the filter element **3** and can be again mounted thereon. Between the housing bottom part **24** and the housing top part **25**, a seal element, for example, in the form of an O-ring, can be provided. The housing top part **25** can comprise quick connect closures **26** of which in FIG. **1** only one is provided with a reference character. The number of quick connect closures **26** is arbitrary. For example, three such quick connect closures **26** are provided which are arranged uniformly distributed around the symmetry axis **4**.

(28) By means of the quick connect closures **26**, the housing top part **25** can be connected detachably to the housing bottom part **24**. For this purpose, engagement sections, for example, in the form of hooks or steps, can be provided at the housing bottom part **24**, in which the quick connect closures **26** engage with form fit for connecting the housing top part **25** to the housing bottom part **24**. A form fit connection is produced by mutual engagement with each other or engagement from behind of at least two connection partners, presently the quick connect closures **26** and the engagement sections. The housing top part **25** comprises furthermore engagement sections which can interact with the positioning elements **22** of the second end disk **10** of the filter element **3** in such a way that the positioning elements **22** engage with form fit the engagement sections of the housing top part **25**. For example, the housing top part **25** is an injection-molded

plastic part.

(29) The housing bottom part **24** is embodied in a cup shape and comprises a cylindrical base section **27** which is constructed with rotational symmetry in relation to the symmetry axis **4**. At the end face, the base section **27** is closed by means of a bottom section **28**. The base section **27** and the bottom section **28** are constructed as one piece, in particular monolithic as one piece. “One piece” or “one part” means presently that the base section **27** and the bottom section **28** form a common component and are not assembled of different individual components. “Monolithic as one piece” means presently that the base section **27** and the bottom section **28** are manufactured throughout of the same material. For example, the housing bottom part **24** is an injection-molded plastic part.

(30) The housing bottom part **24** comprises a fluid inlet **29** which is of a tubular configuration. The fluid inlet **29** is constructed with rotational symmetry in relation to a center or symmetry axis **30**. The symmetry axis **30** is positioned perpendicularly to the symmetry axis **4**. Through the fluid inlet **29**, the fluid L to be purified can be supplied at the raw side to the filter element **3**. Furthermore, the housing bottom part **24** comprises a fluid outlet **31** which is provided at the bottom section **28**. The fluid outlet **31** is tubular and constructed with rotational symmetry in relation to the symmetry axis **4**. Through the fluid outlet **31**, the purified fluid L can be discharged from the filter element **3**.

(31) The fluid outlet **31** extends, beginning at the bottom section **28** of the housing bottom part **24**, outwardly in the direction away from the filter element **3**. Furthermore, as an extension of the fluid outlet **31**, a tubular interface **33** (FIG. **8**) extends into an interior **32** (FIGS. **3** to **6**) of the housing bottom part and interacts with the interface **20** of the filter element **3** in order to seal the filter element **3** in relation to the housing bottom part **24**. The interface **33** is of a tubular configuration and embodied with rotational symmetry in relation to the symmetry axis **4**. The interface **33** can also be referred to as second interface or as filter housing interface.

(32) At the inner side at the interface **33**, this means facing away from the interface **20** of the filter element **3**, a disturbance contour **34** is provided at the interface **33**. The disturbance contour **34** is, for example, embodied as a plurality of grooves extending along the longitudinal direction LR. The disturbance contour **34** prevents that a filter element that does not belong to the filter system **1** can be mounted at the interface **33** which would radially inwardly seal relative to the interface **33**.

(33) The interface **33** extends, as mentioned before, from the bottom section **28** into the interior **32** of the housing bottom part **24**. In this context, the interface **33** comprises a cylindrical seal surface **35** which is constructed with rotational symmetry in relation to the symmetry axis **4** and which interacts with the seal surface **17** of the filter element **3**. In particular, the seal surfaces **17**, **35**, viewed in the radial direction R, are radially compressed with each other.

(34) Viewed along the longitudinal direction LR, a nose or seal rib **36** extending circumferentially in an annular shape about the symmetry axis **4** adjoins the seal surface **35**. The seal rib **36** is suitable to engage with form fit the seal groove **18** of the interface **20**. Viewed in the longitudinal direction LR, a cylindrical centering surface **37** is provided behind the seal rib **36**. The centering surface **37** is suitable to center or to guide the seal surface **17** of the filter element **3** upon installation thereof in the housing bottom part **24** in relation to the symmetry axis **4**. Between the surface **19** and centering surface **37**, a gap **38**, in particular an air gap, is provided.

(35) As illustrated in FIG. **9**, at the bottom section **28** of the housing bottom part **24** disturbance geometries **39** are provided of which in FIG. **9** only one is provided with a reference character. For example, three or five such disturbance geometries **39** are provided which are arranged uniformly distributed around the symmetry axis **4**. The number of disturbance geometries **39** is arbitrary. The disturbance geometries **39** are suitable to engage with form fit the positioning recesses **15** of the first end disk **9**. The disturbance geometries **39** project, beginning at the bottom section **28**, into the interior **32**. The disturbance geometries **39** prevent furthermore the installation of a filter element, without positioning recesses **15** and not matching the filter system, into the housing bottom part **24**.

(36) Viewed from an end surface **40** (FIG. **8**) of the bottom section **28** against which the end

surface **16** rests, each disturbance geometry **39**, viewed along the longitudinal direction LR, has a depth t**39**. In this context, the depth t**39** is larger than the depth t**18** and smaller than the depth t**15**. Each disturbance geometry **39** comprises an end surface **41** which is oriented parallel to the end surface **40** and spaced apart therefrom. The base section **12** of the first end disk **9** comprises an end surface **42**. The end surfaces **41**, **42** are positioned parallel to each other and spaced apart from each other.

(37) In the housing bottom part **24**, furthermore centering geometries **43** are integrally formed of which in FIGS. **2** and **9** only one is provided with a reference character, respectively. For example, three or five such centering geometries **43** can be provided which are arranged uniformly distributed around the symmetry axis **4**. Each centering geometry **43** comprises a top edge **44** which is inclined at a slant relative to the symmetry axis **4**.

(38) The installation of the filter element **3** in the filter housing **2** will be explained in the following with the aid of FIGS. **4** to **6** and **8**. First, the filter element **3** is inserted into the housing bottom part **24** in an insertion direction E which is oriented along the symmetry axis **4**. The insertion direction E is oriented in this context from the second end disk **10** in the direction of the first end disk **9**. The longitudinal direction LR and the insertion direction E can be oppositely oriented. In this context, a pre-centering of the filter element **3** by means of the plate-shaped base section **12** of the first end disk **9** at the centering geometries **43** takes place. By means of the centering geometries **43**, the filter element **3** is centered in relation to the symmetry axis **4** so that the end surface **16** of the positioning and sealing section **14** of the first end disk **9** contacts the end surfaces **41** of the disturbance geometries **39** (FIG. **5**). This means that the interfaces **20**, **33** are not yet in engagement with each other.

(39) From the position illustrated in FIG. **5**, the filter element **3** can now be rotated about the symmetry axis **4** until the disturbance geometries **39** are aligned with the positioning recesses **15** of the positioning and sealing section **14** so that the filter element **3** can be pushed farther into the housing bottom part **24** along the insertion direction E. Upon rotation of the filter element **3** about the symmetry axis **4**, the seal surface **17** of the first end disk **9** is guided at the centering surface **37** of the interface **33** and is centered in relation to the symmetry axis **4**. As soon the disturbance geometries **39** engage the positioning recesses **15**, the positioning and sealing section **14** is elastically deformed such that the seal rib **36** engages with form fit the seal groove **18**. At the same time, the seal surfaces **17**, **35** are radially compressed against each other. The end surfaces **16**, **40** rest against each other. The filter element **3** is mounted in the housing bottom part **24**.

(40) The filter system **1** comprises furthermore a muffler **45** (FIGS. **1** to **3** and **7**) which is attached to the fluid inlet **29**. The muffler **45** is preferably a one-part plastic component, in particular monolithic as one piece. The muffler **45** can be an injection-molded plastic part. The muffler **45** is constructed with rotational symmetry in relation to the symmetry axis **30**. The muffler **45** comprises at the exterior a plurality of fluid guiding ribs **46** which extend parallel to the symmetry axis **30**. The fluid guiding ribs **46** are provided at the exterior at a tubular base body **47** of the muffler **45**. The fluid L to be purified is supplied to the filter element **3** through the muffler **45**.

(41) The base body **47** comprises a truncated cone-shaped inlet **48** as well as an also truncated cone-shaped outlet **49**. The inlet **48** and outlet **49** are in fluid communication with each other. The inlet **48** and outlet **49** are arranged such that the truncated cone-shaped geometries are positioned such that between the inlet **48** and outlet **49** a cross section constriction **51** that is rounded by a rounded portion **50** is provided. The inlet **48** is facing away from the fluid inlet **29**. The outlet **49** is facing the fluid inlet **29**. The inlet **48** and outlet **49** together form thus an hourglass-shaped or trumpet-shaped geometry. The outlet **49** comprises an inflow cross section A of the fluid inlet. The filter medium **5** is provided with inflow via the inflow cross section A.

(42) At the inlet **48**, furthermore an inlet rounded portion **52** is provided which extends circumferentially completely around the symmetry axis **30**. The inlet rounded portion **52** extends circumferentially completely around an inlet opening **53** of the base body **47**. The base body **47**

passes into a tubular fastening section **54**. The fastening section **54** can comprise snap hooks **55** by means of which the muffler **45** is connected to the fluid inlet **29** by form fit. Between the fastening section **54** and base body **47**, a rib **56** extending circumferentially completely around the symmetry axis **30** can be provided. The rib **56** in this context is arranged perpendicularly to the symmetry axis **30**. The rib **56** is received in the fluid inlet **29**. The fluid guiding ribs **46** are provided at an exterior side **57** (FIGS. **1** and **2**) of the base body **47**. The outlet **49** comprises an outlet opening **58**. A diameter of the outlet opening **58** is smaller than a diameter of the inlet opening **53**.
(43) In operation of the filter system **1**, the fluid L to be filtered is sucked in around the inlet rounded portion **52** laterally into the inlet opening **53** and thus into the inlet **48**, as illustrated in FIG. **7** by means of the arrows **59**, **60**. The fluid L flows thus along the fluid guiding ribs **46** which supply the fluid to the inlet **48**. Through the base body **47**, the fluid L flows along an in particular first flow direction SR**1**. The flow direction SR**1** is oriented from the inlet opening **53** in the direction of the filter element **3**. The fluid guiding ribs **46** extend along or parallel to the flow direction SR**1**.
(44) The fluid L flows at the exterior at the base body **47** along the fluid guiding ribs **46** in an in particular second flow direction SR**2**. The flow directions SR**1**, SR**2** are oppositely oriented. The flow direction SR**2** is oriented along the radial direction R. The flow direction SR**1**, on the other hand, is oriented opposite to the radial direction R. The fluid guiding ribs **46** extend also along the flow direction SR**2**.
(45) Immediately upstream of the inlet **48**, a region **61** is provided in which the fluid L substantially has no movement. This means that the fluid L to be filtered is substantially sucked in only along the fluid guiding ribs **46** in the direction of the inlet rounded portion **52** and around the latter into the inlet **48**. The sucked-in fluid L impacts on the filter medium **5**, wherein the stabilization ring **6** prevents a movement of folds of the folded filter medium **5**. In this context, the stabilization ring **6**, viewed along the longitudinal direction LR, is positioned centrally in the inflow cross section A of the fluid inlet **29**.
(46) In particular, the filter medium **5** is protected by means of the stabilization ring **6** from pulsations. In this way, a noise reduction is provided. The stabilization ring **6** in this context is centrally arranged in relation to the muffler **45**. This means the symmetry axis **30** extends preferably centrally through the stabilization ring **6**. The double cone shape of the inlet **48** and of the outlet **49** provides for noise reduction.
REFERENCE CHARACTERS
(47) **1** filter system **2** filter housing **3** filter element **4** symmetry axis **5** filter medium **6** stabilization ring **7** end face **8** end face **9** end disk **10** end disk **11** passage **12** base section **13** cutout **14** positioning and sealing section **15** positioning recess **16** end surface **17** seal surface **18** seal groove **19** surface **20** interface **21** base section **22** positioning element **23** interior **24** housing bottom part **25** housing top part **26** quick connect closure **27** base section **28** bottom section **29** fluid inlet **30** symmetry axis **31** fluid outlet **32** interior **33** interface **34** disturbance contour **35** seal surface **36** seal rib **37** centering surface **38** gap **39** disturbance geometry **40** end surface **41** end surface **42** end surface **43** centering geometry **44** top edge **45** muffler **46** fluid guiding ribs **47** base body **48** inlet **49** outlet **50** rounded portion **51** cross section constriction **52** inlet rounded portion **53** inlet opening **54** fastening section **55** snap hook **56** rib **57** exterior side **58** outlet opening **59** arrow **60** arrow **61** region A inflow cross section E insertion direction L fluid LR longitudinal direction R radial direction RL clean side RO raw side SR**1** flow direction SR**2** flow direction t**15** depth t**17** depth t**18** depth t**20** depth t**39** depth

## Claims

1. A filter element for a filter system, the filter element comprising: a filter medium; a first end disk connected to the filter medium and a second end disk connected to the filter medium, wherein the

filter medium is arranged between the first end disk and the second end disk; wherein the first end disk comprises a positioning and sealing section facing away from the filter medium; wherein the positioning and sealing section comprises externally arranged positioning recesses configured to engage disturbance geometries of a filter housing of the filter system with form fit and to position the filter element circumferentially in relation to the filter housing; wherein the positioning and sealing section comprises an inner side and an interface arranged at the inner side, wherein the interface is configured to seal the filter element radially in relation to the filter housing and to position the filter element axially in relation to the filter housing; wherein the interface, viewed along a longitudinal direction of the filter element oriented from the first end disk toward the second end disk, extends farther into the positioning and sealing section than the externally arranged positioning recesses; wherein the interface comprises an annular seal groove surrounding circumferentially completely a symmetry axis of the filter element; wherein the annular seal groove is configured to engage a seal rib of the filter housing with form fit and to position the filter element axially in relation to the filter housing; wherein the interface comprises a circumferentially extending seal surface configured to seal the filter element radially in relation to the filter housing; wherein the annular seal groove, viewed along the longitudinal direction, is arranged behind the circumferentially extending seal surface; wherein the interface further comprises a circumferentially extending surface positioned such that the annular seal groove, viewed along the longitudinal direction, is arranged between the circumferentially extending seal surface and the circumferentially extending surface; wherein the circumferentially, extending surface, viewed along the longitudinal direction, extends farther into the positioning and sealing section than the externally arranged positioning recesses; and wherein the circumferentially extending surface comprises a larger diameter than the circumferentially extending seal surface.

2. The filter element according to claim 1, wherein the first end disk is open and the second end disk is closed.

3. The filter element according to claim 1, wherein the externally arranged positioning recesses are arranged uniformly distributed or non-uniformly distributed around a circumference of the filter element.

4. A filter system comprising: a filter housing comprising disturbance geometries and a seal surface; a filter element comprising: a filter medium; a first end disk connected to the filter medium and a second end disk connected to the filter medium, wherein the filter medium is arranged between the first end disk and the second end disk; wherein the first end disk comprises a positioning and sealing section facing away from the filter medium; wherein the positioning and sealing section comprises externally arranged positioning recesses engaging the disturbance geometries of the filter housing with form fit and positioning the filter element circumferentially in relation to the filter housing; wherein the positioning and sealing section further comprises an inner side and an interface arranged at the inner side, wherein the interface is configured to seal the filter element radially in relation to the filter housing and to position the filter element axially in relation to the filter housing; wherein the interface, viewed along a longitudinal direction of the filter element oriented from the first end disk toward the second end disk, extends farther into the positioning and sealing section than the externally arranged positioning recesses; wherein the seal surface of the filter housing, at least in sections, radially compresses the interface of the filter element, and wherein a respective end surface of the disturbance geometries of the filter housing, viewed along the longitudinal direction, is arranged behind the seal surface of the filter housing.

5. The filter system according to claim 4, wherein the positioning and sealing section of the filter element, when installing the filter element in the filter housing, first contacts with an end face thereof the disturbance geometries, and wherein the disturbance geometries and the externally arranged positioning recesses are brought into form fit engagement with each other by rotating the filter element in relation to the filter housing.

6. The filter system according to claim 5, wherein a seal surface of the interface of the filter

element, when installing the filter element in the filter housing, contacts the seal surface of the filter housing not until the disturbance geometries and the externally arranged positioning recesses are engaged with each other with form fit.
7. The filter system according to claim 6, wherein the disturbance geometries, viewed along the longitudinal direction, comprise a larger depth than the seal surface of the interface of the filter element.
8. The filter system according to claim 4, wherein the filter housing comprises a seal rib, wherein the interface of the filter element comprises an annular seal groove surrounding circumferentially completely a symmetry axis of the filter element, wherein the annular seal groove engages the seal rib of the filter housing with form fit and positions the filter element axially in relation to the filter housing, and wherein the seal rib of the filter housing, viewed along the longitudinal direction, is arranged behind the seal surface of the filter housing.
9. The filter system according to claim 8, wherein the filter housing comprises a centering surface, wherein the centering surface, viewed along the longitudinal direction, is arranged behind the seal rib of the filter housing.
10. The filter system according to claim 9, wherein a circumferentially extending gap is provided between the centering surface of the filter housing and the interface of the filter element when the filter element is in an installed position in the filter housing.
11. The filter system according to claim 4, wherein the disturbance geometries are arranged uniformly distributed or non-uniformly distributed around a circumference of the filter housing.
12. A filter element for a filter system, the filter element comprising: a filter medium; a first end disk connected to the filter medium and a second end disk connected to the filter medium, wherein the filter medium is arranged between the first end disk and the second end disk; wherein the first end disk comprises a positioning and sealing section facing away from the filter medium; wherein the positioning and sealing section comprises externally arranged positioning recesses configured to engage disturbance geometries of a filter housing of the filter system with form fit and to position the filter element circumferentially in relation to the filter housing; wherein the positioning and sealing section comprises an inner side and an interface arranged at the inner side, wherein the interface is configured to seal the filter element radially in relation to the filter housing and to position the filter element axially in relation to the filter housing; wherein the interface, viewed along a longitudinal direction of the filter element oriented from the first end disk toward the second end disk, extends farther into the positioning and sealing section than the externally arranged positioning recesses; wherein the interface comprises an annular seal groove surrounding circumferentially completely a symmetry axis of the filter element; wherein the annular seal groove is configured to engage a seal rib of the filter housing with form fit and to position the filter element axially in relation to the filter housing; wherein the interface comprises a circumferentially extending seal surface configured to seal the filter element radially in relation to the filter housing; wherein the annular seal groove, viewed along the longitudinal direction, is arranged behind the circumferentially extending seal surface; and wherein the externally arranged positioning recesses, viewed along the longitudinal direction, project farther into the positioning and sealing section than the circumferentially extending seal surface.
13. The filter element according to claim 12, wherein the first end disk is open and the second end disk is closed.
14. The filter element according to claim 12, wherein the externally arranged positioning recesses are arranged uniformly distributed or non-uniformly distributed around a circumference of the filter element.
15. The filter element according to claim 12, wherein the interface comprises a circumferentially extending surface positioned such that the circumferentially extending annular seal groove, viewed along the longitudinal direction, is arranged between the circumferentially extending seal surface and the circumferentially extending surface, and wherein the circumferentially extending surface,

viewed along the longitudinal direction, extends farther into the positioning and sealing section than the externally arranged positioning recesses.
16. The filter element according to claim 15, wherein the circumferentially extending surface comprises a larger diameter than the circumferentially extending seal surface.